



RE20 – Biofuel

The entire process of producing biofuel is demonstrated with the Armfield Bio-Fuel kit in the form of student experiments.

The suitcase contains all necessary parts and components and can be used from any location. Our first step is resource selection and fermentation. The resulting mash is then distilled with the custom-built condenser and the resulting ethanol will be characterized.

The Armfield Biofuel kit does not only cover bio ethanol production but also the generation of biodiesel through transesterification of fats.

PRODUCT CODE: RE20

Categories: Educational, RE Series - Renewable Energy, RE20 Biofuel

Tags: Alcoholmeter , Bio Ethanol , Biodiesel , Biofuel , Calcium Hydroxide , Cooking Oil , Distillation , Ethanol-Fuel Cell , FAME , Fermented Mash , Mash I , RE20 , Renewable Energy , Transesterification Of Fats

DESCRIPTION

The entire process of producing biofuel is demonstrated with the Armfield Bio-Fuel kit in the form of student experiments.

The suitcase contains all necessary parts and components and can be used from any location. Our first step is resource selection and fermentation. The resulting mash is then distilled with the custom-built condenser and the resulting ethanol will be characterized.

Lastly, the produced bio fuel needs to be converted into usable energy – for example into electricity with the provided ethanol fuel cell.

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TECHNICAL SPECIFICATIONS

- 1 x Potentiometer module
- 1 x Motor module without gear
- 1 x Ethanol fuel cell module
- 1 x Plug with hose

- 1 x Yeast
- 1 x Chain clamp
- 2 x Digital multi-meter
- 3 x Test lead black 25 cm
- 2 x Test lead red 25 cm
- 4 x Bump on transparent 5.0 mm height X 11.1mm diameter
- 1 x Propeller
- 1 x Laboratory thermometer
- 1 x Distilling head, 2 cores 75°, NS 19/26
- 1 x Condenser
- 1 x Alcoholmeter
- 1 x Erlenmeyer flask 1000ml
- 1 x Airlock
- 1 x Rubber stopper
- 1 x Areometer
- 1 x Beaker 250ml
- 3 x Test tubes
- 1 x Grip stopper
- 3 x Pasteur pipette
- 1 x Measuring cylinder 100ml
- 1 x Syringe 2ml
- 1 x Silicone ring
- 1 x Info sheet initial startup
- 1 x Universal stand clamp
- 1 x Stand rod 60cm, M10
- 2 x Double clamp
- 1 x Stand base plate
- 1 x Aluminium case BioFuel
- 1 x BioFuel Manual

FEATURES & BENEFITS

- Includes an Ethanol-fuel cell for the generation of electrical energy out of biofuel
- Production of biofuel displayed in experiments for students
- Interdisciplinary experiments for chemistry, physics and biology
- All additional devices already included – usable without laboratory equipment

Experimental Content

Biofuel experiments:

- Production of biodiesel (FAME) from fats and oils
- Extraction of fats from groceries or oil plants
- Differences between cooking oil and FAME
- Fermentation on the basis of different sugars
- Proof of CO₂ by a calcium hydroxide solution
- Link between reaction speed and temperature
- Distillation of fermented mash
- Working principle of the fuel cell



- Recording the characteristic curve of a 15% solution
- Temperature dependence of the power
Concentration dependence of the power and the characteristic curve

Related Curriculums

- Renewable Energies
- Chemical Engineering
- Environmental Engineering

Other products in the series

RE10: Advanced Photovoltaic Energy

RE12: Advanced Wind Energy

RE14: Advanced Fuel Cell Technology

RE16: Advanced Thermal Energy

RE24: Advanced Battery Technology

Operational Conditions

- Storage Temperature: -10°C to +70°C
- Operating temperature range: +10°C to +50°C
- Operating relative humidity range: 0 to 95%, non-condensing

Requirements

Electrical supply: 110-230V AC 50-60Hz

- Level and stable work surface
- 1 x Supplementary BioFuel
- 1 x Bunsen burner (Not supplied by Armfield)

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

- **Volume:** 0.038m³
- **Gross weigh:** 10Kg

Dimensions

TRAY

- **Length:** 0.640m
- **Width:** 0.165m
- **Height:** 0.370m

Order Codes



RE20: Bio-Fuel
RE21: Biofuel Clamp Set

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